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Step 23 - Architecture Review and Final Assessment

Reference Note

This chapter summarizes the ARM Generic Timer Architecture covered throughout this study series.

The goal is to connect all major concepts into a single architecture view and provide a final assessment for self-review and interview preparation.

Learning Objectives

After completing this chapter, you should be able to:

- Explain the complete Generic Timer architecture
- Describe the relationship between counters and timers
- Explain timer virtualization
- Describe timer interrupt delivery
- Understand timer ownership across exception levels
- Solve basic timer calculations
- Answer common timer interview questions

1. Complete Architecture Overview

Core Architecture

Component	Purpose
System Counter	Master source of time
Physical Counter	Physical view of time
Virtual Counter	Virtualized view of time
Physical Timer	Event generation from physical time

Virtual Timer	Event generation from virtual time
Hypervisor Timer	EL2 timer events
Secure Timer	Secure-world timer events

2. End-to-End Timer Flow

Step	Operation
1	System Counter advances
2	Counter value observed
3	Timer compare logic evaluates
4	Compare condition becomes true
5	Timer expires
6	PPI asserted
7	GIC marks interrupt pending
8	CPU accepts interrupt
9	ISR executes

3. Counter Architecture Review

Shared Resource

Component	Scope
System Counter	Entire System

There is only:

One System Counter

Counter Views

Register	Purpose

CNTPCT_EL0	Physical Time
CNTVCT_EL0	Virtual Time

Both originate from:

System Counter

4. Timer Architecture Review

Physical Timer

Register	Purpose
CNTP_CVAL_EL0	Absolute deadline
CNTP_TVAL_EL0	Relative interval
CNTP_CTL_EL0	Timer control

Virtual Timer

Register	Purpose
CNTV_CVAL_EL0	Virtual deadline
CNTV_TVAL_EL0	Virtual interval
CNTV_CTL_EL0	Timer control

5. Important Formulas

Time to Ticks

Ticks = Time × Frequency

Example

Parameter	Value
Time	1 ms
Frequency	24 MHz

Result:

24,000 ticks

Ticks to Time

Time = Ticks / Frequency

Example

Parameter	Value
Ticks	48,000
Frequency	24 MHz

Result:

2 ms

6. Relative vs Absolute Time

Relative Time

Uses:

TVAL

Meaning:

Fire X ticks from now

Absolute Time

Uses:

CVAL

Meaning:

Fire when counter reaches value Y

Recommended Approach

Application	Preferred Method
One-Shot Delay	TVAL or CVAL
Periodic Timer	CVAL
Accurate Scheduling	CVAL

7. Virtualization Review

Key Formula

CNTVCT
=
CNTPCT - CNTVOFF_EL2

Components

Component	Purpose
CNTPCT	Physical Time
CNTVOFF_EL2	Offset
CNTVCT	Virtual Time

Benefit

One System Counter
Multiple Virtual Timelines

8. Exception Level Ownership Review

Resource	Typical Owner
Physical Timer	EL1
Virtual Timer	Guest EL1
Hypervisor Timer	EL2
Secure Timer	EL3

9. Timer Interrupt Review

Timer Expiry Condition

Current Counter >= Compare Value

Control Bits

Field	Purpose
ENABLE	Enable Timer
IMASK	Mask Interrupt Output
ISTATUS	Expiry Status

Important Facts

Statement	True?
IMASK stops timer counting	No

ISTATUS means timer expired	Yes
Timer expiry guarantees ISR	No
Timer expiry raises interrupt request	Yes

10. PPI Review

Generic Timer PPIs

Timer	Typical PPI
Hypervisor Timer	PPI 26
Virtual Timer	PPI 27
Secure Timer	PPI 29
Physical Timer	PPI 30

Key Concept

Shared Counter

Private Timers

Private Interrupts

11. GIC Review

Interrupt States

State	Meaning
Inactive	No request
Pending	Waiting for service
Active	ISR executing
Active + Pending	Re-triggered

Interrupt Path

Timer
↓
PPI
↓
GIC
↓
CPU
↓
ISR

12. SMP Review

Shared Components

Component
System Counter
Counter Frequency

Per-Core Components

Component
Physical Timer
Virtual Timer
PPIs

Benefit

Consistent timestamps
Easy task migration
Shared timeline

13. DVFS Review

CPU Frequency

Can change:

500 MHz
1 GHz
2 GHz

System Counter Frequency

Remains fixed.

Example:

24 MHz

Key Principle

CPU Speed Changes
Time Does Not

14. Final Architecture Summary

Component	Key Role
System Counter	Measures time
Counter	Provides current time
Timer	Generates events
PPI	Interrupt source
GIC	Interrupt delivery

ISR	Software processing
CNTVOFF_EL2	Virtualization offset

Practice Problems

Question 1

At:

24 MHz

how many ticks occur in:

10 ms?

Answer

24,000,000 × 0.01
=
240,000 ticks

Question 2

Current Counter:

1,000,000

Delay:

24,000 ticks

What CVAL should be programmed?

Answer

1,024,000

Question 3

If:

CNTPCT = 1,000,000
CNTVOFF = 250,000

What is CNTVCT?

Answer

750,000